GENERATIVE AGENT SIMULATIONS OF HUMAN BEHAVIOR

A DISSERTATION
SUBMITTED TO THE DEPARTMENT OF COMPUTER SCIENCE
AND THE COMMITTEE ON GRADUATE STUDIES
OF STANFORD UNIVERSITY
IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF
DOCTOR OF PHILOSOPHY

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August 2025

This dissertation is online at: https://purl.stanford.edu/jm164ch6237

I certify that I have read this dissertation and that, in my opinion, it is fully adequate in scope and quality as a dissertation for the degree of Doctor of Philosophy.

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Abstract

Human institutions embed assumptions about how people perceive, decide, and act; when those assumptions are incomplete, well-intentioned systems can produce unintended effects at scale. This dissertation argues that simulations powered by generative, language-model–based agents can serve as a new scientific instrument for probing such assumptions, revealing emergent dynamics, and exploring counterfactual designs before deployment. I introduce a general architecture that transforms large language models into coherent computational agents through three capabilities—memory (structured episodic records), reflection (periodic synthesis of goals), and planning (multi-timescale plans that guide and revise action). Complementing this scaffold, I develop methods for constructing agents from rich, individual-level data (e.g., interviews) and a measurement framework that evaluates alignment against ground truth on people's attitudes and behaviors.

Applying these components, the dissertation presents an interactive sandbox demonstrating coherent individual behavior and emergent social phenomena, and a population simulation of 1,000 agents representing a cross-section of U.S. adults used to assess fidelity across attitudinal, behavioral, and diffusion tasks and to analyze where alignment succeeds or fails. Case studies show how generative-agent simulations can inform platform design, content recommendation, and public policy, with social simulacra illustrating how to prototype communities and surface failure modes prior to real-world deployment. Taken together, the work traces a path from capability to credibility, positioning generative-agent simulations as instruments that complement empirical observation to challenge assumptions, generate testable hypotheses, and support wiser design and governance of social systems.

Acknowledgements

This dissertation is dedicated to the people who have made my past five years as a researcher meaningful.

- **Advisers—Michael Bernstein and Percy Liang.** Thank you for the support and guidance that will stay with me in everything I do. You taught me to dream ambitiously and gave me the tools to pursue those dreams. Your insight, rigor, and care for the craft—and for the people you work with—shaped every step of this dissertation.

- **Mentors—Meredith Ringel Morris and Karrie Karahalios.** Thank you for trusting me and supporting me over the years, not only in scholarship but in learning how to be a good colleague and community member.

- **Social science mentor—Robb Willer.** For modeling rigor in experimental design, for generous mentorship, and for the good humor and groovy vibes, always.

- **Early champions—James Landay, Jeff Hancock, Danae Metaxa, and Mary Wootters.** Thank you for taking an early bet on me and helping me find a place to grow as an early-stage scholar.

- **Collaborators.** This work was made possible—and far more joyful—by collaborators whose creativity and care sharpened the ideas: Carolyn Zou, Joseph O'Brien, Lindsay Popowski, Aaron Shaw, Mako Hill, and Carrie Cai.

- **Friends and colleagues.** The Stanford HCI and NLP communities made everyday work rich and entertaining. Thanks to my officemates—Mitchell

Gordon, Michelle Lam, Beleicia Bullock, and Jordan Troutman—and to my cohorts Rishi Bommasani, Lisa Li, and Rohith Kuditipudi for conversations that pushed my thinking. I am also very grateful to Mia Tang for helping me wrangle notoriously finicky LaTeX tables in this dissertation.

- **Family.** To my parents, Ji Yeon Kim and Jeong-Jun Park: your love, patience, and example are the foundation I value most. Thank you for everything.

This PhD unfolded during a period of extraordinary change in our field. GPT-3 became widely available in my first year, catalyzing a wave of new ideas. Working at the intersection of HCI and AI, I set out to explore meaningful applications for these models—work that eventually led me to generative agents and simulations. I am grateful to have done this research during such a formative moment, and humbled by the responsibility we share to shape this technology for the better.

Lastly, a note on place. The setting of *Smallville*, where the generative agents live, is an amalgam of real spaces that fed and sustained me as an undergraduate and graduate student. Thank you to everyone in those places for the meals, conversations, and community that nourished this work over the years.

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Chapter 1

Introduction

Human societies are complex systems. The institutions we design, the platforms we build, and the policies we enact are built on assumptions about how people perceive, decide, and act. When those assumptions are wrong, or even just incomplete, well-intentioned systems can yield unintended consequences at scale. This dissertation argues that simulation, and in particular simulations powered by generative, language-model–based agents, can serve as a new kind of scientific instrument for probing those assumptions, challenging prevailing explanations, and exploring counterfactual designs before they are deployed in the world.

To motivate this claim, consider a canonical example from the study of complex social systems: Thomas Schelling's 1971 model of residential segregation (Schelling, 1971, 1978). In a simple grid world populated by two types of agents, each agent is content to live in a mixed neighborhood and requires only a minimal share of similar neighbors—say, 30%—to be satisfied. Agents who fall below this threshold move to the nearest acceptable location. Despite these weak individual preferences, the collective dynamics produce strong segregation. The striking lesson was not that actors are bigots; rather, that weak, local preferences can yield large-scale, unintended patterns. That insight reshaped how scholars and policymakers reasoned about segregation, shifting attention from overt animus to the emergent consequences of individually reasonable choices.

Schelling's example illustrates two broader points. First, simulations can expose how micro-level rules produce macro-level outcomes that defy intuition. Second, the force of those insights depends on the behavioral assumptions we encode. For decades, computational social science relied on highly stylized agents: a handful of fixed parameters stood in for human cognition, motivation, and social context. These models yielded elegant theory and valuable intuition, but their simplicity made it difficult to generalize across domains or populations, and easy to overlook the contingency and diversity that characterize real human behavior.

Recent advances in generative AI offer a new opportunity. Large language models (LLMs) are trained on broad corpora that reflect many facets of contemporary life—narratives, plans, conversations, and social practices. When leveraged appropriately, they can generate plausible descriptions of goals, beliefs, and actions across a wide range of backgrounds and settings. This suggests a path toward general-purpose computational agents that preserve the flexibility of natural language while remaining programmable, inspectable, and composable into large-scale simulations.

But two challenges stand in the way: First, long-term coherence. People do not act as isolated, stateless predictors of the next token in a conversation. We accumulate experiences; we recall, reinterpret, and integrate them; we plan and revise. Off-the-shelf language models, even with expanding context windows, cannot naively reason over the full arc of an agent's lived experience. Without additional machinery, generated behavior drifts, forgets, and fragments. For simulations to be useful as instruments of inquiry, agents must remember, reflect, and plan over extended horizons, doing so efficiently and in ways that scale to many agents.

Second, accuracy and validation. Believability is not enough. If simulations are to inform consequential decisions in science, policy, and design, they must be empirically anchored. We need principled methods to construct agents that represent diverse, real individuals and to evaluate whether simulated attitudes and behaviors match ground-truth measures—and to know where they do not. Moreover, naïve conditioning on demographics risks stereotyping: a description like "30-year-old Asian graduate student" should not, by itself, license generic or biased inferences about daily life. The core methodological question is how to build and test agents that are faithful to individuals and populations rather than caricatures of them.

This dissertation tackles these challenges by developing architectures, methods, and measurement frameworks for generative agent simulations of human behavior. The central thesis is:

> With the right agent architectures, generative language models can be transformed into coherent, empirically grounded computational agents, enabling simulations that both reveal emergent social dynamics and provide decision-relevant evidence for policy and design.

1.1 From Stylized Agents to Generative Agents

The work proceeds in a tradition that spans cognitive architectures, agent-based models, social AI, and virtual worlds. Prior models afforded clarity through parsimony: agents had few states and followed simple rules (Bruch and Atwell, 2015; Epstein and Axtell, 1996). The cost was rigidity. They struggled to capture the idiosyncratic goals and contextual reasoning that make human behavior contingent.

Generative agents invert this trade-off. By operating in natural language and being conditioned on rich prompts and experience, they can flexibly represent goals, beliefs, social ties, and routines—without hand-coding domain-specific rules. To make this practical, I introduce an agent architecture with three core capabilities:

1. **Memory.** Agents maintain a structured record of experiences ("episodic memory") that includes observations, actions, and social interactions. Rather than carrying the entire record into every decision, agents retrieve only the most salient memories, using learned or heuristic relevance measures tailored to the current situation.

2. **Reflection.** Periodically, agents distill higher-level inferences from their episodic memories: evolving goals, social impressions, and personal preferences. These reflective summaries stabilize behavior across time and allow agents to update their self-models in light of new evidence.

3. **Planning.** Agents generate and revise plans at multiple time scales (e.g., daily routines, near-term tasks), grounding actions in both current context and